
Pin Map

Pin Name	Pin	Pin Name	Pin
1CLR	14	1D	15
1CLK	16	1PRE	17
1Q	18	1Qnot	19
2Qnot	1	2Q	2
2PRE	3	2CLK	4
2D	5	2CLR	6

Global Parameters

Parameter	Value
VCC Pin	7
GND Pin	20
VCC Voltage	2.5, 3.3, 4.0, 4.5, 5
Input Low	0.75, 0.99, 1.2, 1.35, 1.5
Input High	1.75, 2.31, 2.8, 3.15, 3.5
Output Low	0.3, 0.3, 0.3, 0.3, 0.3
Output High	2.2, 3.0, 3.6, 4.1, 4.7

Tests: **IC FAIL 2/4**

D Flip-Flop 1: FAIL

Inputs				VCC	Outputs/Results			
1PRE	1CLR	1D	1CLK		1Q		1Qnot	
L	H	X	X	2.5	H	2.37 (H)	L	0.0 (L)
				3.3		3.15 (H)		0.0 (L)
				4.0		3.81 (H)		0.0 (L)
				4.5		4.43 (H)		0.0 (L)
				5		4.85 (H)		0.0 (L)

Inputs				VCC	Outputs/Results			
1PRE	1CLR	1D	1CLK		1Q		1Qnot	
H	L	X	X	2.5	L	0.0 (L)	H	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.43 (H)
				5		0.0 (L)		4.86 (H)
L	L	X	X	2.5	H	2.37 (H)	H	2.37 (H)
				3.3		3.15 (H)		3.15 (H)
				4.0		3.81 (H)		3.81 (H)
				4.5		4.43 (H)		4.43 (H)
				5		4.85 (H)		4.86 (H)
H	H	H	R	2.5	H	2.37 (H)	L	0.0 (L)
				3.3		3.15 (H)		0.0 (L)
				4.0		3.81 (H)		0.0 (L)
				4.5		4.43 (H)		0.0 (L)
				5		4.85 (H)		0.0 (L)
H	H	L	R	2.5	L	0.0 (L)	H	0.0 (L)
				3.3		0.0 (L)		0.0 (L)
				4.0		0.0 (L)		0.0 (L)
				4.5		0.0 (L)		0.0 (L)
				5		0.0 (L)		0.0 (L)

D Flip-Flop 2: PASS

Inputs				VCC	Outputs/Results			
2PRE	2CLR	2D	2CLK		2Q		2Qnot	
L	H	X	X	2.5	H	2.36 (H)	L	0.0 (L)
				3.3		3.15 (H)		0.0 (L)
				4.0		3.81 (H)		0.0 (L)
				4.5		4.43 (H)		0.0 (L)
				5		4.85 (H)		0.0 (L)
H	L	X	X	2.5	L	0.0 (L)	H	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.43 (H)
				5		0.0 (L)		4.86 (H)
L	L	X	X	2.5	H	2.37 (H)	H	2.37 (H)
				3.3		3.15 (H)		3.15 (H)
				4.0		3.81 (H)		3.81 (H)
				4.5		4.43 (H)		4.43 (H)
				5		4.86 (H)		4.86 (H)
H	H	H	R	2.5	H	2.37 (H)	L	0.0 (L)
				3.3		3.15 (H)		0.0 (L)
				4.0		3.81 (H)		0.0 (L)
				4.5		4.43 (H)		0.0 (L)
				5		4.86 (H)		0.0 (L)

Inputs				VCC	Outputs/Results			
2PRE	2CLR	2D	2CLK		2Q		2Qnot	
H	H	L	R	2.5	L	0.0 (L)	H	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.43 (H)
				5		0.0 (L)		4.86 (H)

No Change -> H: PASS

Inputs				VCC	Outputs/Results							
1PRE, 2PRE	1CLR, 2CLR	1D, 2D	1CLK, 2CLK		1Q		2Q		1Qnot		2Qnot	
H	H	H	R	2.5	H	2.37 (H)	H	2.37 (H)	L	0.0 (L)	L	0.0 (L)
				3.3		3.15 (H)		3.15 (H)		0.0 (L)		0.0 (L)
				4.0		3.81 (H)		3.81 (H)		0.0 (L)		0.0 (L)
				4.5		4.43 (H)		4.43 (H)		0.0 (L)		0.0 (L)
				5		4.86 (H)		4.86 (H)		0.0 (L)		0.0 (L)
H	H	X	L	2.5	H	2.37 (H)	H	2.37 (H)	L	0.0 (L)	L	0.0 (L)
				3.3		3.15 (H)		3.15 (H)		0.0 (L)		0.0 (L)
				4.0		3.81 (H)		3.81 (H)		0.0 (L)		0.0 (L)
				4.5		4.43 (H)		4.43 (H)		0.0 (L)		0.0 (L)
				5		4.85 (H)		4.85 (H)		0.0 (L)		0.0 (L)

No Change -> L: FAIL

Inputs				VCC	Outputs/Results							
1PRE, 2PRE	1CLR, 2CLR	1D, 2D	1CLK, 2CLK		1Q		2Q		1Qnot		2Qnot	
H	H	L	R	2.5	L	0.0 (L)	L	0.0 (L)	H	0.0 (L)	H	2.37 (H)
				3.3		0.0 (L)		0.0 (L)		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		0.0 (L)		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		0.0 (L)		0.0 (L)		4.43 (H)
				5		0.0 (L)		0.0 (L)		0.0 (L)		4.85 (H)
H	H	X	L	2.5	L	0.0 (L)	L	0.0 (L)	H	0.0 (L)	H	2.37 (H)
				3.3		0.0 (L)		0.0 (L)		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		0.0 (L)		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		0.0 (L)		0.0 (L)		4.43 (H)
				5		0.0 (L)		0.0 (L)		0.0 (L)		4.85 (H)